Configuration options and examples

Astra Trident

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Table of Contents

Configuration options and examples . . . 1
Backend configuration options . . . 1
Minimal configuration examples . . . 6
Examples of backends with virtual storage pools . . . 8
Map backends to StorageClasses . . . 12

Configuration options and examples

Learn about how to create and use ONTAP NAS drivers with your Astra Trident installation. This section provides backend configuration examples and details about how to map backends to StorageClasses.

Backend configuration options

See the following table for the backend configuration options:

Parameter	Description	Default
`version`		Always 1
`storageDriverName`	Name of the storage driver	“ontap-nas”, “ontap-nas-economy”, “ontap-nas-flexgroup”, “ontap-san”, “ontap-san-economy”
`backendName`	Custom name or the storage backend	Driver name + “_” + dataLIF
`managementLIF`	IP address of a cluster or SVM management LIF	“10.0.0.1”, “[2001:1234:abcd::fefe]”
`dataLIF`	IP address of protocol LIF. Use square brackets for IPv6. Cannot be updated after you set it	Derived by the SVM unless specified
`autoExportPolicy`	Enable automatic export policy creation and updating [Boolean]	false
`autoExportCIDRs`	List of CIDRs to filter Kubernetes’ node IPs against when `autoExportPolicy` is enabled	[“0.0.0.0/0”, “::/0”]`
`labels`	Set of arbitrary JSON-formatted labels to apply on volumes	“”
`clientCertificate`	Base64-encoded value of client certificate. Used for certificate-based auth	“”
`clientPrivateKey`	Base64-encoded value of client private key. Used for certificate-based auth	“”
`trustedCACertificate`	Base64-encoded value of trusted CA certificate. Optional. Used for certificate-based auth	“”
`username`	Username to connect to the cluster/SVM. Used for credential-based auth	
`password`	Password to connect to the cluster/SVM. Used for credential-based auth	

Parameter	Description	Default
`svm`	Storage virtual machine to use	Derived if an SVM `managementLIF` is specified
`igroupName`	Name of the igroup for SAN volumes to use	“trident-<backend-UUID>”
`storagePrefix`	Prefix used when provisioning new volumes in the SVM. Cannot be updated after you set it	“trident”
`limitAggregateUsage`	Fail provisioning if usage is above this percentage. **Does not apply to Amazon FSx for ONTAP**	“” (not enforced by default)
`limitVolumeSize`	Fail provisioning if requested volume size is above this value for the economy driver.	“” (not enforced by default)
`lunsPerFlexvol`	Maximum LUNs per Flexvol, must be in range [50, 200]	“100”
`debugTraceFlags`	Debug flags to use when troubleshooting. Example, {“api”:false, “method”:true}	null
`nfsMountOptions`	Comma-separated list of NFS mount options	“”
`qtreesPerFlexvol`	Maximum Qtrees per FlexVol, must be in range [50, 300]	“200”
`useREST`	Boolean parameter to use ONTAP REST APIs. **Tech preview**	false

`useREST` is provided as a **tech preview** that is recommended for test environments and not for production workloads. When set to `true`, Astra Trident will use ONTAP REST APIs to communicate with the backend. This feature requires ONTAP 9.8 and later. In addition, the ONTAP login role used must have access to the `ontap` application. This is satisfied by the pre-defined `vsadmin` and `cluster-admin` roles.

To communicate with the ONTAP cluster, you should provide the authentication parameters. This could be the username/password to a security login or an installed certificate.

If you are using an Amazon FSx for NetApp ONTAP backend, do not specify the `limitAggregateUsage` parameter. The `fsxadmin` and `vsadmin` roles provided by Amazon FSx for NetApp ONTAP do not contain the required access permissions to retrieve aggregate usage and limit it through Astra Trident.

Do not use `debugTraceFlags` unless you are troubleshooting and require a detailed log dump.

When creating a backend, remember that the `dataLIF` and `storagePrefix` cannot be modified after creation. To update these parameters, you will need to create a new backend.

A fully-qualified domain name (FQDN) can be specified for the managementLIF option. A FQDN may also be specified for the dataLIF option, in which case the FQDN will be used for the NFS mount operations. This way you can create a round-robin DNS to load-balance across multiple data LIFs.

managementLIF for all ONTAP drivers can also be set to IPv6 addresses. Make sure to install Astra Trident with the --use-ipv6 flag. Care must be taken to define the managementLIF IPv6 address within square brackets.

> When using IPv6 addresses, make sure managementLIF and dataLIF (if included in your backend definition) are defined within square brackets, such as [28e8:d9fb:a825:b7bf:69a8:d02f:9e7b:3555]. If dataLIF is not provided, Astra Trident will fetch the IPv6 data LIFs from the SVM.

Using the autoExportPolicy and autoExportCIDRs options, CSI Trident can manage export policies automatically. This is supported for all ontap-nas-* drivers.

For the ontap-nas-economy driver, the limitVolumeSize option will also restrict the maximum size of the volumes it manages for qtrees and LUNs, and the qtreesPerFlexvol option allows customizing the maximum number of qtrees per FlexVol.

The nfsMountOptions parameter can be used to specify mount options. The mount options for Kubernetes persistent volumes are normally specified in storage classes, but if no mount options are specified in a storage class, Astra Trident will fall back to using the mount options specified in the storage backend's configuration file. If no mount options are specified in either the storage class or the configuration file, then Astra Trident will not set any mount options on an associated persistent volume.

> Astra Trident sets provisioning labels in the "Comments" field of all volumes created using ontap-nas and ontap-nas-flexgroup. Based on the driver used, the comments are set on the FlexVol (ontap-nas) or FlexGroup (ontap-nas-flexgroup). Astra Trident will copy all labels present on a storage pool to the storage volume at the time it is provisioned. Storage administrators can define labels per storage pool and group all volumes created in a storage pool. This provides a convenient way of differentiating volumes based on a set of customizable labels that are provided in the backend configuration.

Backend configuration options for provisioning volumes

You can control how each volume is provisioned by default using these options in a special section of the configuration. For an example, see the configuration examples below.

Parameter	Description	Default
spaceAllocation	Space-allocation for LUNs	"true"
spaceReserve	Space reservation mode; "none" (thin) or "volume" (thick)	"none"
snapshotPolicy	Snapshot policy to use	"none"
qosPolicy	QoS policy group to assign for volumes created. Choose one of qosPolicy or adaptiveQosPolicy per storage pool/backend	""

Parameter	Description	Default
`adaptiveQosPolicy`	Adaptive QoS policy group to assign for volumes created. Choose one of qosPolicy or adaptiveQosPolicy per storage pool/backend. Not supported by ontap-nas-economy.	“”
`snapshotReserve`	Percentage of volume reserved for snapshots “0”	If `snapshotPolicy` is “none”, else “”
`splitOnClone`	Split a clone from its parent upon creation	“false”
`encryption`	Enable NetApp volume encryption	“false”
`securityStyle`	Security style for new volumes	“unix”
`tieringPolicy`	Tiering policy to use “none”	“snapshot-only” for pre-ONTAP 9.5 SVM-DR configuration
unixPermissions	Mode for new volumes	“777”
snapshotDir	Controls visibility of the `.snapshot` directory	“false”
exportPolicy	Export policy to use	“default”
securityStyle	Security style for new volumes	“unix”

> Using QoS policy groups with Astra Trident requires ONTAP 9.8 or later. It is recommended to use a non-shared QoS policy group and ensure the policy group is applied to each constituent individually. A shared QoS policy group will enforce the ceiling for the total throughput of all workloads.

Here’s an example with defaults defined:

```
{
  "version": 1,
  "storageDriverName": "ontap-nas",
  "backendName": "customBackendName",
  "managementLIF": "10.0.0.1",
  "dataLIF": "10.0.0.2",
  "labels": {"k8scluster": "dev1", "backend": "dev1-nasbackend"},
  "svm": "trident_svm",
  "username": "cluster-admin",
  "password": "password",
  "limitAggregateUsage": "80%",
  "limitVolumeSize": "50Gi",
  "nfsMountOptions": "nfsvers=4",
  "debugTraceFlags": {"api":false, "method":true},
  "defaults": {
    "spaceReserve": "volume",
    "qosPolicy": "premium",
    "exportPolicy": "myk8scluster",
    "snapshotPolicy": "default",
    "snapshotReserve": "10"
  }
}
```

For `ontap-nas` and `ontap-nas-flexgroups`, Astra Trident now uses a new calculation to ensure that the FlexVol is sized correctly with the snapshotReserve percentage and PVC. When the user requests a PVC, Astra Trident creates the original FlexVol with more space by using the new calculation. This calculation ensures that the user receives the writable space they requested for in the PVC, and not lesser space than what they requested. Before v21.07, when the user requests a PVC (for example, 5GiB), with the snapshotReserve to 50 percent, they get only 2.5GiB of writeable space. This is because what the user requested for is the whole volume and `snapshotReserve` is a percentage of that. With Trident 21.07, what the user requests for is the writeable space and Astra Trident defines the `snapshotReserve` number as the percentage of the whole volume. This does not apply to `ontap-nas-economy`. See the following example to see how this works:

The calculation is as follows:

```
Total volume size = (PVC requested size) / (1 - (snapshotReserve
percentage) / 100)
```

For snapshotReserve = 50%, and PVC request = 5GiB, the total volume size is 2/.5 = 10GiB and the available size is 5GiB, which is what the user requested in the PVC request. The `volume show` command should show results similar to this example:

```
Vserver   Volume       Aggregate    State      Type       Size  Available Used%
--------- ------------ ------------ ---------- ---- ---------- ---------- -----
                 _pvc_89f1c156_3801_4de4_9f9d_034d54c395f4
                                      online     RW         10GB     5.00GB    0%
                 _pvc_e8372153_9ad9_474a_951a_08ae15e1c0ba
                                      online     RW          1GB    511.8MB    0%
2 entries were displayed.
```

Existing backends from previous installs will provision volumes as explained above when upgrading Astra Trident. For volumes that you created before upgrading, you should resize their volumes for the change to be observed. For example, a 2GiB PVC with `snapshotReserve=50` earlier resulted in a volume that provides 1GiB of writable space. Resizing the volume to 3GiB, for example, provides the application with 3GiB of writable space on a 6 GiB volume.

Minimal configuration examples

The following examples show basic configurations that leave most parameters to default. This is the easiest way to define a backend.

> If you are using Amazon FSx on NetApp ONTAP with Trident, the recommendation is to specify DNS names for LIFs instead of IP addresses.

`ontap-nas` driver with certificate-based authentication

This is a minimal backend configuration example. `clientCertificate`, `clientPrivateKey`, and `trustedCACertificate` (optional, if using trusted CA) are populated in `backend.json` and take the base64-encoded values of the client certificate, private key, and trusted CA certificate, respectively.

```
{
  "version": 1,
  "backendName": "DefaultNASBackend",
  "storageDriverName": "ontap-nas",
  "managementLIF": "10.0.0.1",
  "dataLIF": "10.0.0.15",
  "svm": "nfs_svm",
  "clientCertificate": "ZXR0ZXJwYXB...ICMgJ3BhcGVyc2",
  "clientPrivateKey": "vciwKIyAgZG...0cnksIGRlc2NyaX",
  "trustedCACertificate": "zcyBbaG...b3Igb3duIGNsYXNz",
  "storagePrefix": "myPrefix_"
}
```

`ontap-nas` driver with auto export policy

This example shows you how you can instruct Astra Trident to use dynamic export policies to create and manage the export policy automatically. This works the same for the `ontap-nas-economy` and `ontap-nas-flexgroup` drivers.

```
{
    "version": 1,
    "storageDriverName": "ontap-nas",
    "managementLIF": "10.0.0.1",
    "dataLIF": "10.0.0.2",
    "svm": "svm_nfs",
    "labels": {"k8scluster": "test-cluster-east-1a", "backend": "test1-
nasbackend"},
    "autoExportPolicy": true,
    "autoExportCIDRs": ["10.0.0.0/24"],
    "username": "admin",
    "password": "secret",
    "nfsMountOptions": "nfsvers=4",
}
```

`ontap-nas-flexgroup` driver

{
"version": 1,
"storageDriverName": "ontap-nas-flexgroup",
"managementLIF": "10.0.0.1",
"dataLIF": "10.0.0.2",
"labels": {"k8scluster": "test-cluster-east-1b", "backend": "test1-ontap-cluster"},
"svm": "svm_nfs",
"username": "vsadmin",
"password": "secret",
}

`ontap-nas` driver with IPv6

```
{
 "version": 1,
 "storageDriverName": "ontap-nas",
 "backendName": "nas_ipv6_backend",
 "managementLIF": "[5c5d:5edf:8f:7657:bef8:109b:1b41:d491]",
 "labels": {"k8scluster": "test-cluster-east-1a", "backend": "test1-ontap-
ipv6"},
 "svm": "nas_ipv6_svm",
 "username": "vsadmin",
 "password": "netapp123"
}
```

`ontap-nas-economy` driver

```
{
    "version": 1,
    "storageDriverName": "ontap-nas-economy",
    "managementLIF": "10.0.0.1",
    "dataLIF": "10.0.0.2",
    "svm": "svm_nfs",
    "username": "vsadmin",
    "password": "secret"
}
```

Examples of backends with virtual storage pools

In the sample backend definition file shown below, specific defaults are set for all storage pools, such as `spaceReserve` at none, `spaceAllocation` at false, and `encryption` at false. The virtual storage pools are defined in the storage section.

In this example, some of the storage pool sets their own `spaceReserve`, `spaceAllocation`, and `encryption` values, and some pools overwrite the default values set above.

`ontap-nas` driver

```
{
    {
    "version": 1,
    "storageDriverName": "ontap-nas",
    "managementLIF": "10.0.0.1",
    "dataLIF": "10.0.0.2",
    "svm": "svm_nfs",
    "username": "admin",
    "password": "secret",
    "nfsMountOptions": "nfsvers=4",

    "defaults": {
          "spaceReserve": "none",
          "encryption": "false",
          "qosPolicy": "standard"
    },
    "labels":{"store":"nas_store", "k8scluster": "prod-cluster-1"},
    "region": "us_east_1",
    "storage": [
        {
            "labels":{"app":"msoffice", "cost":"100"},
            "zone":"us_east_1a",
            "defaults": {
                "spaceReserve": "volume",
```

```
                "encryption": "true",
                "unixPermissions": "0755",
                "adaptiveQosPolicy": "adaptive-premium"
            }
        },
        {
            "labels":{"app":"slack", "cost":"75"},
            "zone":"us_east_1b",
            "defaults": {
                "spaceReserve": "none",
                "encryption": "true",
                "unixPermissions": "0755"
            }
        },
        {
            "labels":{"app":"wordpress", "cost":"50"},
            "zone":"us_east_1c",
            "defaults": {
                "spaceReserve": "none",
                "encryption": "true",
                "unixPermissions": "0775"
            }
        },
        {
            "labels":{"app":"mysqldb", "cost":"25"},
            "zone":"us_east_1d",
            "defaults": {
                "spaceReserve": "volume",
                "encryption": "false",
                "unixPermissions": "0775"
            }
        }
    ]
}
```

`ontap-nas-flexgroup` driver

```
{
    "version": 1,
    "storageDriverName": "ontap-nas-flexgroup",
    "managementLIF": "10.0.0.1",
    "dataLIF": "10.0.0.2",
    "svm": "svm_nfs",
    "username": "vsadmin",
    "password": "secret",
```

```
    "defaults": {
          "spaceReserve": "none",
          "encryption": "false"
    },
    "labels":{"store":"flexgroup_store", "k8scluster": "prod-cluster-1"},
    "region": "us_east_1",
    "storage": [
        {
            "labels":{"protection":"gold", "creditpoints":"50000"},
            "zone":"us_east_1a",
            "defaults": {
                "spaceReserve": "volume",
                "encryption": "true",
                "unixPermissions": "0755"
            }
        },
        {
            "labels":{"protection":"gold", "creditpoints":"30000"},
            "zone":"us_east_1b",
            "defaults": {
                "spaceReserve": "none",
                "encryption": "true",
                "unixPermissions": "0755"
            }
        },
        {
            "labels":{"protection":"silver", "creditpoints":"20000"},
            "zone":"us_east_1c",
            "defaults": {
                "spaceReserve": "none",
                "encryption": "true",
                "unixPermissions": "0775"
            }
        },
        {
            "labels":{"protection":"bronze", "creditpoints":"10000"},
            "zone":"us_east_1d",
            "defaults": {
                "spaceReserve": "volume",
                "encryption": "false",
                "unixPermissions": "0775"
            }
        }
    ]
}
```

`ontap-nas-economy` driver

```
{
    "version": 1,
    "storageDriverName": "ontap-nas-economy",
    "managementLIF": "10.0.0.1",
    "dataLIF": "10.0.0.2",
    "svm": "svm_nfs",
    "username": "vsadmin",
    "password": "secret",

    "defaults": {
          "spaceReserve": "none",
          "encryption": "false"
    },
    "labels":{"store":"nas_economy_store"},
    "region": "us_east_1",
    "storage": [
        {
            "labels":{"department":"finance", "creditpoints":"6000"},
            "zone":"us_east_1a",
            "defaults": {
                "spaceReserve": "volume",
                "encryption": "true",
                "unixPermissions": "0755"
            }
        },
        {
            "labels":{"department":"legal", "creditpoints":"5000"},
            "zone":"us_east_1b",
            "defaults": {
                "spaceReserve": "none",
                "encryption": "true",
                "unixPermissions": "0755"
            }
        },
        {
            "labels":{"department":"engineering", "creditpoints":"3000"},
            "zone":"us_east_1c",
            "defaults": {
                "spaceReserve": "none",
                "encryption": "true",
                "unixPermissions": "0775"
            }
        },
        {
```

```
            "labels":{"department":"humanresource",
"creditpoints":"2000"},
            "zone":"us_east_1d",
            "defaults": {
                "spaceReserve": "volume",
                "encryption": "false",
                "unixPermissions": "0775"
            }
        }
    ]
}
```

Map backends to StorageClasses

The following StorageClass definitions refer to the above virtual storage pools. Using the `parameters.selector` field, each StorageClass calls out which virtual pool(s) can be used to host a volume. The volume will have the aspects defined in the chosen virtual pool.

- The first StorageClass (`protection-gold`) will map to the first, second virtual storage pool in the `ontap-nas-flexgroup` backend and the first virtual storage pool in the `ontap-san` backend. These are the only pool offering gold level protection.
- The second StorageClass (`protection-not-gold`) will map to the third, fourth virtual storage pool in `ontap-nas-flexgroup` backend and the second, third virtual storage pool in `ontap-san` backend. These are the only pools offering protection level other than gold.
- The third StorageClass (`app-mysqldb`) will map to the fourth virtual storage pool in `ontap-nas` backend and the third virtual storage pool in `ontap-san-economy` backend. These are the only pools offering storage pool configuration for mysqldb type app.
- The fourth StorageClass (`protection-silver-creditpoints-20k`) will map to the third virtual storage pool in `ontap-nas-flexgroup` backend and the second virtual storage pool in `ontap-san` backend. These are the only pools offering gold-level protection at 20000 creditpoints.
- The fifth StorageClass (`creditpoints-5k`) will map to the second virtual storage pool in `ontap-nas-economy` backend and the third virtual storage pool in `ontap-san` backend. These are the only pool offerings at 5000 creditpoints.

Astra Trident will decide which virtual storage pool is selected and will ensure the storage requirement is met.

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: protection-gold
provisioner: netapp.io/trident
parameters:
  selector: "protection=gold"
  fsType: "ext4"
---
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: protection-not-gold
provisioner: netapp.io/trident
parameters:
  selector: "protection!=gold"
  fsType: "ext4"
---
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: app-mysqldb
provisioner: netapp.io/trident
parameters:
  selector: "app=mysqldb"
  fsType: "ext4"
---
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: protection-silver-creditpoints-20k
provisioner: netapp.io/trident
parameters:
  selector: "protection=silver; creditpoints=20000"
  fsType: "ext4"
---
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: creditpoints-5k
provisioner: netapp.io/trident
parameters:
  selector: "creditpoints=5000"
  fsType: "ext4"
```

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